

C792

Data Modeling and Database Management Systems

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A nurse educator at a school of nursing with 400 students is designing a database to track clinical rotations. Each student has a name, student ID, and cohort year. The school's mascot is a hawk. In the ERD, what is cohort year?

- A. A relationship.
- B. An attribute of the Student entity.
- C. The primary key.
- D. An entity.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2.

A clinic is designing a patient table. Last name is always filled in, and the clinic has 8,000 patients, many sharing surnames. The clinic staff wear blue scrubs. Which column is the best primary key?

- A. Last name.
- B. First name plus last name.
- C. Phone number.
- D. A system-assigned patient ID.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

In a hospital medication database, the Administration table stores the NurseID for each dose given. A database administrator who has worked there 11 years tries to delete a nurse from the Nurse table who has 300 recorded administrations. Referential integrity is enforced with no cascade rule. What happens?

- A. The delete is rejected.
- B. The delete succeeds because of the administrator's long service.
- C. The nurse and all 300 administration rows are deleted.
- D. The nurse is deleted and the 300 rows keep a NurseID that no longer exists.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4. [TRANSFER — reversed stem / different variable]

A designer wants every Visit row to link to exactly one existing Patient. Which constraint belongs on Visit.PatientID?

- A. A NOT NULL foreign key referencing Patient.PatientID.
- B. A CHECK that PatientID is greater than zero.
- C. Make Visit.PatientID the primary key of Visit.
- D. A UNIQUE constraint on Visit.PatientID.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

A unit tracks which nurses hold which certifications. One nurse can hold many certifications, and each certification is held by many nurses. The unit has 45 nurses and 9 certification types. How should this be modeled?

- A. A NurseID column in the Certification table.
- B. A junction table, such as NurseCertification, with foreign keys to Nurse and Certification.
- C. Nine certification columns in the Nurse table.
- D. One CertificationID column in the Nurse table.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6. [TRANSFER — reversed stem / different variable]

A junction table links Patient and Allergy. What kind of relationship does it resolve?

- A. One-to-many from Allergy to Patient.
- B. A recursive (self-referencing) relationship.
- C. One-to-one.
- D. Many-to-many.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

A table Visit(VisitID, PatientID, PatientName, VisitDate) has VisitID as its primary key. A clinic analyst notes the table has 50,000 rows and loads slowly on Mondays. PatientName depends on PatientID, not directly on VisitID. Which normal form does this violate?

- A. First normal form.
- B. Second normal form.
- C. Third normal form (transitive dependency).
- D. None; slow loading is a hardware issue.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8. [TRANSFER — reversed stem / different variable]

A table keyed on (StudentID, CourseID) also stores StudentName, which depends only on StudentID. Which normal form is violated?

- A.** First normal form.
- B.** Second normal form.
- C.** None; composite keys cannot have dependency problems.
- D.** Third normal form.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

A quality nurse at a hospital with an award-winning cafeteria has a Patient table with 50 patients and a Visit table with 45 visits. Thirty patients have at least one visit, and every visit belongs to a valid patient. How many rows does Patient LEFT JOIN Visit return?

- A.** 50
- B.** 95
- C.** 65
- D.** 45

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10. [TRANSFER — reversed stem / different variable]

An analyst needs the nursing units whose average length of stay is greater than 5 days. Which clause filters on the average?

- A.** ORDER BY
- B.** WHERE
- C.** GROUP BY alone
- D.** HAVING

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

A home health agency with a new logo stores patient temperatures in a text field. Some entries read 98.6, others “normal,” and others 37C. Which design change best improves data integrity?

- A. Keep everything as text so no entry is ever rejected.
- B. Make the text field longer.
- C. Use a numeric data type with one defined unit and a CHECK constraint on a plausible range.
- D. Add a comments column for explanations.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

A nursing research database at a university hospital with two helipads holds identifiable patient data. Student researchers only need counts by diagnosis. What is the best access design?

- A. Give students a role with read access only to a de-identified, aggregated view.
- B. Deny all access, since research may never use patient data.
- C. Give full read access to base tables after students sign a confidentiality pledge.
- D. Give students administrator rights so they do not need help.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer B Concept 1

A nurse educator at a school of nursing ~~with 400 students~~ is designing a database to track clinical rotations. Each student has a name, student ID, and cohort year. ~~The school's mascot is a hawk.~~ In the ERD, what is cohort year?

Target word: ERD element type **Decides:** Cohort year describes a student.

A	Confusion set	Relationships connect entities, such as Student to Rotation.
B	Correct	It is a descriptive detail about each student.
C	Dropped element	Drops uniqueness; many students share a cohort year.
D	Adjacent category	Student is the entity; cohort year describes it.

2. Answer D Concept 2

A clinic is designing a patient table. Last name is always filled in, and the clinic has 8,000 patients, many sharing surnames. ~~The clinic staff wear blue scrubs.~~ Which column is the best primary key?

Target word: best key **Decides:** Must be unique, not null, and stable.

A	Dropped element	Always present but not unique.
B	Confusion set	Still not unique in 8,000 patients.
C	Adjacent category	Numbers change and can be shared by a household.
D	Correct	Guaranteed unique, never null, and does not change.

3. Answer A Concept 3

In a hospital medication database, the Administration table stores the NurseID for each dose given. A database administrator ~~who has worked there 11 years~~ tries to delete a nurse from the Nurse table who has 300 recorded administrations. Referential integrity is enforced with no cascade rule. What happens?

Target word: result of the delete **Decides:** Child rows still reference that NurseID.

A	Correct	Referential integrity blocks deleting a parent that child rows reference.
B	Camouflage	Seniority does not override a database constraint.
C	Adjacent category	That is cascade delete, which was not set.
D	Dropped element	Drops the integrity rule that prevents orphans.

4. Answer A Concept 3 • transfer

A designer wants every Visit row to link to exactly one existing Patient. Which constraint belongs on Visit.PatientID?

Target word: reverse: goal given, choose the constraint **Decides:** Must match an existing parent row.

A	Correct	Ensures each visit points to a real patient.
B	Dropped element	Checks the format but not that the patient exists.
C	Confusion set	Would allow only one visit per patient.
D	Adjacent category	Also limits each patient to one visit and does not check existence.

5. Answer B Concept 4

A unit tracks which nurses hold which certifications. One nurse can hold many certifications, and each certification is held by many nurses. ~~The unit has 45 nurses and 9 certification types.~~ How should this be modeled?

Target word: model for M:N **Decides:** Many on both sides.

A	Inverted operation	Lets each certification belong to only one nurse.
B	Correct	A junction table resolves many-to-many.
C	Wrong stem number	A repeating group that breaks 1NF and fails when types are added.
D	Adjacent category	That models one-to-many; a nurse could hold only one.

6. Answer D Concept 4 • transfer

A junction table links Patient and Allergy. What kind of relationship does it resolve?

Target word: reverse: structure given, name the relationship **Decides:** Junction tables resolve many-to-many.

A	Inverted operation	Would limit each patient to one allergy.
B	Adjacent category	Recursive relationships link a table to itself.
C	Confusion set	One-to-one needs no junction table.
D	Correct	Patients have many allergies; each allergy occurs in many patients.

7. Answer C Concept 5

A table Visit(VisitID, PatientID, PatientName, VisitDate) has VisitID as its primary key. A clinic analyst notes the table ~~has 50,000 rows and loads slowly on Mondays.~~ PatientName depends on PatientID, not directly on VisitID. Which normal form does this violate?

Target word: violated normal form **Decides:** Single-column key with a non-key → non-key dependency.

A	Confusion set	Values are atomic; 1NF is met.
B	Adjacent category	2NF concerns partial dependency on a composite key; this key is one column.
C	Correct	PatientName depends on PatientID, a non-key column.
D	Camouflage	Performance does not decide the normal form.

8. Answer B Concept 5 • transfer

A table keyed on (StudentID, CourseID) also stores StudentName, which depends only on StudentID. Which normal form is violated?

Target word: composite-key case **Decides:** Dependency on part of a composite key.

A	Adjacent category	Values are still atomic.
B	Correct	A partial dependency on part of the key violates 2NF.
C	Absolute language	Composite keys are exactly where partial dependencies occur.
D	Confusion set	3NF is about non-key to non-key dependencies.

9. Answer C Concept 6

A quality nurse ~~at a hospital with an award-winning cafeteria~~ has a Patient table with 50 patients and a Visit table with 45 visits. Thirty patients have at least one visit, and every visit belongs to a valid patient. How many rows does Patient LEFT JOIN Visit return?

Target word: LEFT JOIN row count **Decides:** Every visit row plus one row for each patient without visits.

A	Wrong stem number	Assumes one row per patient, but patients with several visits repeat.
B	Inverted operation	Adds the two table sizes instead of matching rows.
C	Correct	45 matched visit rows + 20 patients with no visits (50 – 30) = 65.
D	Adjacent category	That is the INNER JOIN count; it drops patients without visits.

10. Answer D Concept 6 · transfer

An analyst needs the nursing units whose average length of stay is greater than 5 days. Which clause filters on the average?

Target word: filter on an aggregate **Decides:** The condition applies after grouping.

A	Adjacent category	Sorts results but does not filter.
B	Confusion set	WHERE filters rows before aggregates exist.
C	Dropped element	Groups the rows but drops the filter condition.
D	Correct	HAVING filters groups after aggregation.

11. Answer C Concept 7

A home health agency ~~with a new logo~~ stores patient temperatures in a text field. Some entries read 98.6, others “normal,” and others 37C. Which design change best improves data integrity?

Target word: integrity fix **Decides:** Mixed formats and units in one field.

A	Absolute language	Rejecting invalid data is the purpose of constraints.
B	Dropped element	Drops the root problem: free text allows any entry.
C	Correct	Blocks words, mixed units, and impossible values at entry.
D	Adjacent category	Adds notes but still allows bad values.

12. Answer A Concept 8

A nursing research database at a university hospital ~~with two helipads~~ holds identifiable patient data. Student researchers only need counts by diagnosis. What is the best access design?

Target word: access design **Decides:** Students need aggregates, not identifiers.

A	Correct	Least privilege: exactly the data needed, nothing more.
B	Absolute language	De-identified or approved research use is allowed.
C	Adjacent category	A pledge is an administrative step, not a technical limit on exposure.
D	Inverted operation	Maximum rather than minimum privilege.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	6	
Wrong stem number	A figure copied from the prompt that is not the answer.	2	
Adjacent category	The right idea in the wrong bucket.	12	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	4	
Absolute language	Always, never, guarantee — where judgment applies.	3	
Confusion set	Every option is real; the test is exact role-matching.	7	
Camouflage	Answers a stem fact that was never the question.	2	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Entities, attributes, relationships	An entity is a thing tracked (a table); an attribute describes it (a column); a relationship links entities.	Calling a descriptive field an entity.	A noun with its own list of details vs. a detail about that noun.
2	Primary keys	A primary key must be unique, never null, and stable; a system-assigned ID is usually best.	Using names or phone numbers as keys.	Many people share a value in the candidate column.
3	Foreign keys and referential integrity	A foreign key must match an existing parent key; with integrity enforced and no cascade, deleting a referenced parent is blocked.	Thinking deletes leave orphan rows or cascade by default.	Deleting a parent row that child rows point to.
4	Resolving many-to-many	A many-to-many relationship needs a junction table holding foreign keys to both sides.	Adding repeating columns or a single foreign key.	"One can have many, and each is held by many."
5	Normalization	1NF: atomic values; 2NF: no partial dependency on part of a composite key; 3NF: no transitive dependency on a non-key column.	Calling any dependency problem a 2NF violation.	Single-column key → 3NF; composite key → 2NF.
6	Joins and aggregates in SQL	INNER JOIN returns only matches; LEFT JOIN keeps every left row; HAVING filters groups after aggregation, WHERE filters rows before.	Expecting a LEFT JOIN to return one row per left record, or using WHERE on an average.	"Include patients with no visits" or "groups whose average..."
7	Data types and constraints	Choose data types and CHECK or domain constraints that make invalid entries impossible.	Storing everything as text so nothing is rejected.	Mixed units or words in a numeric field.
8	Database security by least privilege	Grant roles only the access they need, often through views that hide identifiers or return aggregates.	Relying on pledges instead of technical limits.	Users need summaries but the base tables hold identifiers.

Course scope (public description): This graduate course is designed to engage the student in planning, analyzing, and designing a relational database management system (DBMS) for use by nurse administrators, clinicians, educators, and informaticists. This experience will provide the knowledge needed to advocate for nursing informatics needs within the field of healthcare.